

δS of Balls Is a Linear Functional of Local Energy. The Kernel Is Not Uniquely CHM.

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Abstract

The claim “entanglement gives linearized gravity” needs a linear map from a source to δS of balls. This note builds that map for a free lattice fermion.

One Gaussian potential, 512 balls of radius 2 on $N = 12$. $\delta S(\varepsilon)$ versus $\delta S(2\varepsilon)$ has Pearson $1 - 7 \times 10^{-9}$. δS versus the CHM-weighted energy $\sum (R^2 - r^2) \delta e$ has $\rho = -0.986$.

A linear $R - r$ weight is slightly better ($R = 0.165$ vs 0.168). An auditor on $N = 10$ still tracks CHM ($\rho = -0.975$) but *flat* beats CHM. A smooth source does not separate the kernels.

So: a linear functional, yes. Einstein’s kernel, no. Not $8\pi G$. Not FGHMV in AdS. Not de Sitter.

1 What this is

The first-law half of Faulkner–Guica–Hartman–Myers–Van Raamsdonk, as a lattice measurement, not a bulk Einstein equation. n_{occ} was stable. The locked $\max |\delta S| / \langle S \rangle$ floor failed because S is the area and δS is first order; linearity (C1) is the real signal check and it passed.

Admission. *I ran the test that can speak to a linear functional. I did not promote a 0.02 residual gap into uniqueness, and I did not write $G_{\mu\nu} = 8\pi G T_{\mu\nu}$.*